Independent, outcome-blind evaluation of AlphaGenome Atlas

A Frozen evaluation workflow

B Exact-context evaluation and frozen roles

MPRA cell	Atlas ontology	role	directional n
A549	EFO:0001086	descriptive	2,249
HepG2	EFO:0001187	confirmatory	13,045
K562	EFO:0002067	confirmatory	15,809
HCT116	EFO:0002824	confirmatory	5,866
SK-N-SH	EFO:0003072	confirmatory	13,762

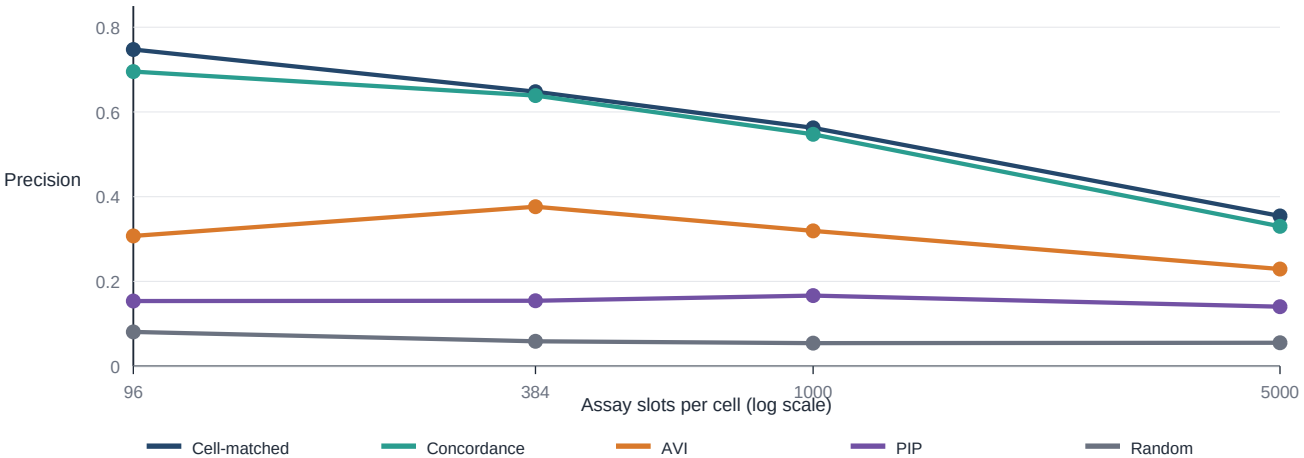
C Integrity and independence checks

Atlas completeness: **246,674 / 246,674**	Prediction inspection: **273 / 273 checks**
Frozen-result inspection: **25 / 25 checks**	Manuscript analysis inspection: **100 / 100 checks**
Gate outcomes in confirmation: **0**	Duplicate variant-cell rows: **0**

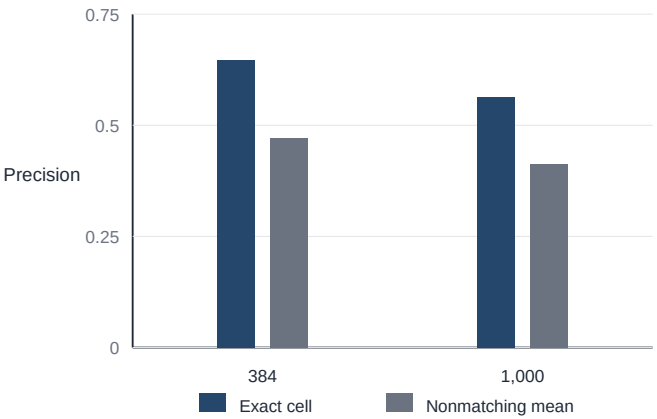
All policy definitions, thresholds and outcomes were frozen before held-out outcome access.

Cell context improves AlphaGenome regulatory-variant prioritization

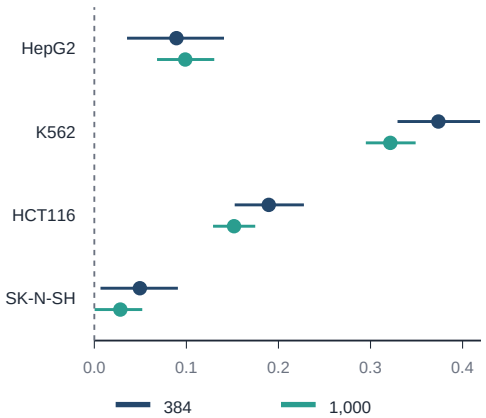
A Macro emVar precision across fixed assay budgets



B Exact-cell versus nonmatching-context mean



C Exact-cell benefit by outcome cell



D Additional confirmed variants and genomic blocks

384 slots

+417 emVars

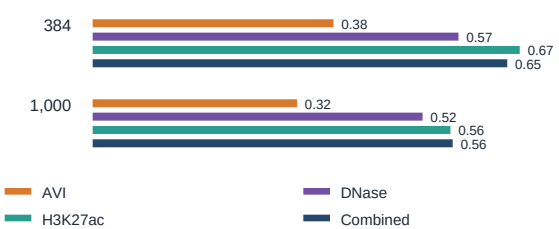
+367 hit-containing 1 Mb blocks

1,000 slots

+973 emVars

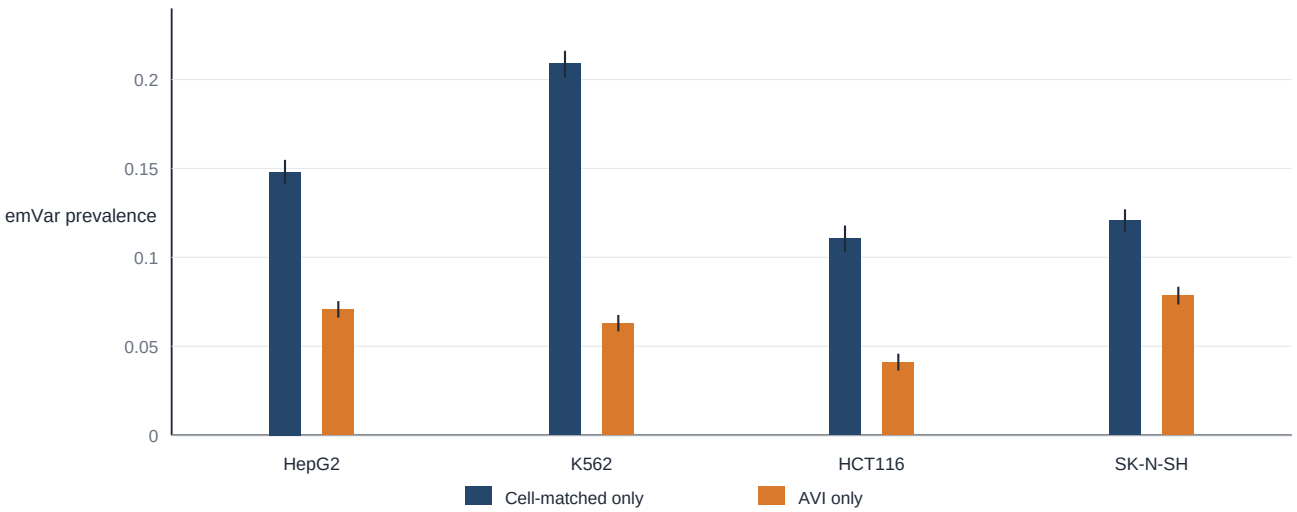
+674 hit-containing 1 Mb blocks

E Post hoc single-track macro precision

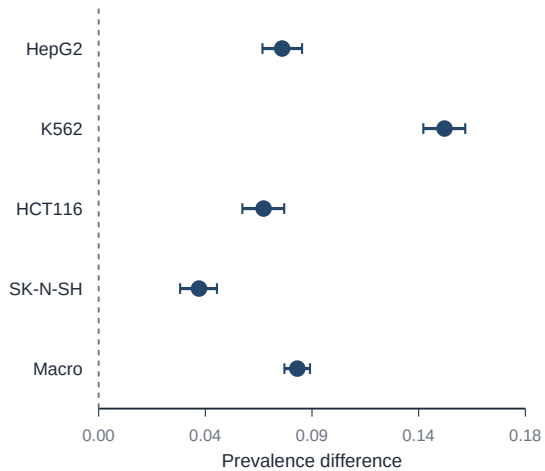


Disagreement is informative; concordance leaves ranking unimproved

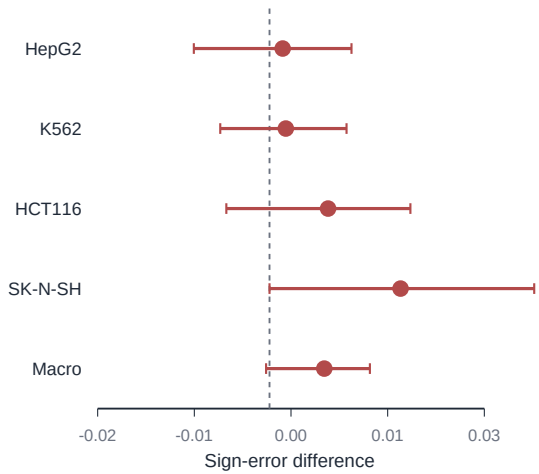
A Top-10% rank discordance separates MPRA yield



B Cell-matched-only minus AVI-only prevalence



C Prespecified concordance hypothesis

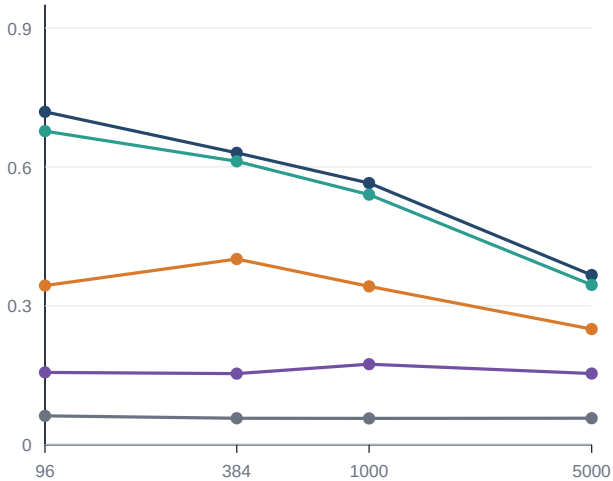


D Agreement marks emVar enrichment

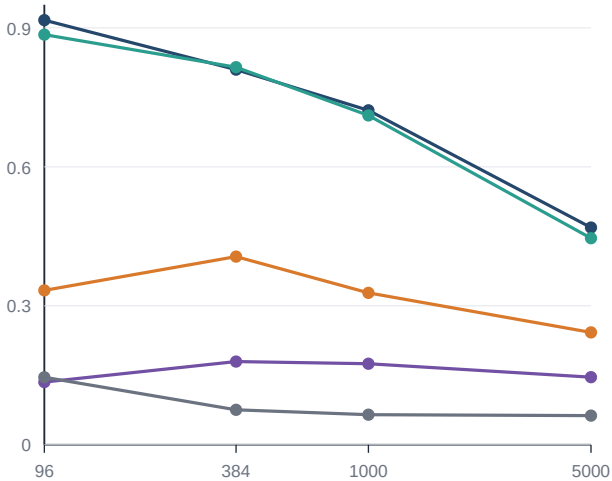
HepG2	K562	HCT116	SK-N-SH
1.78x	1.82x	1.74x	1.41x

Supplementary Figure S1. Frozen policy performance by cell

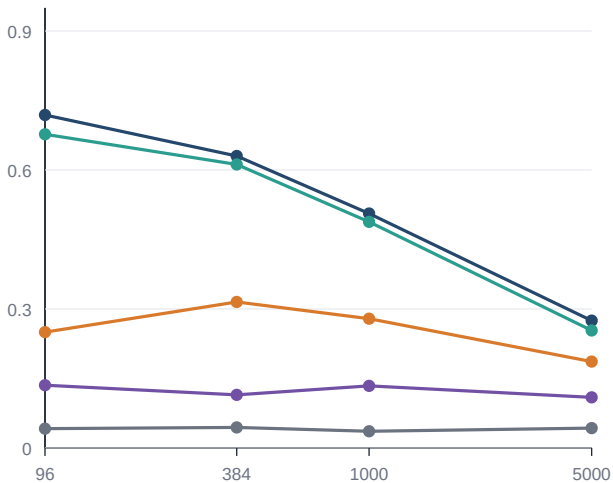
A HepG2



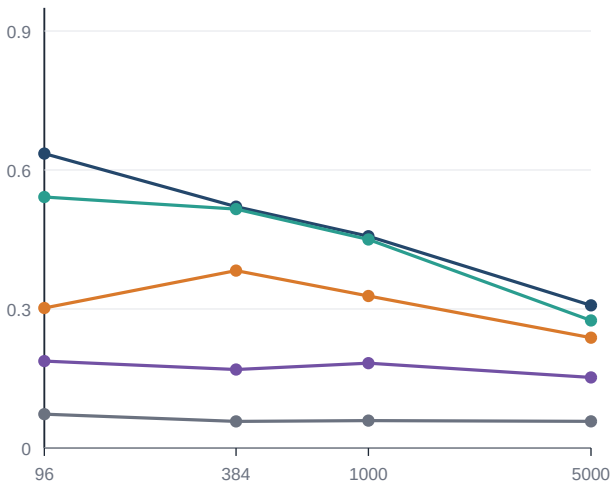
B K562



C HCT116

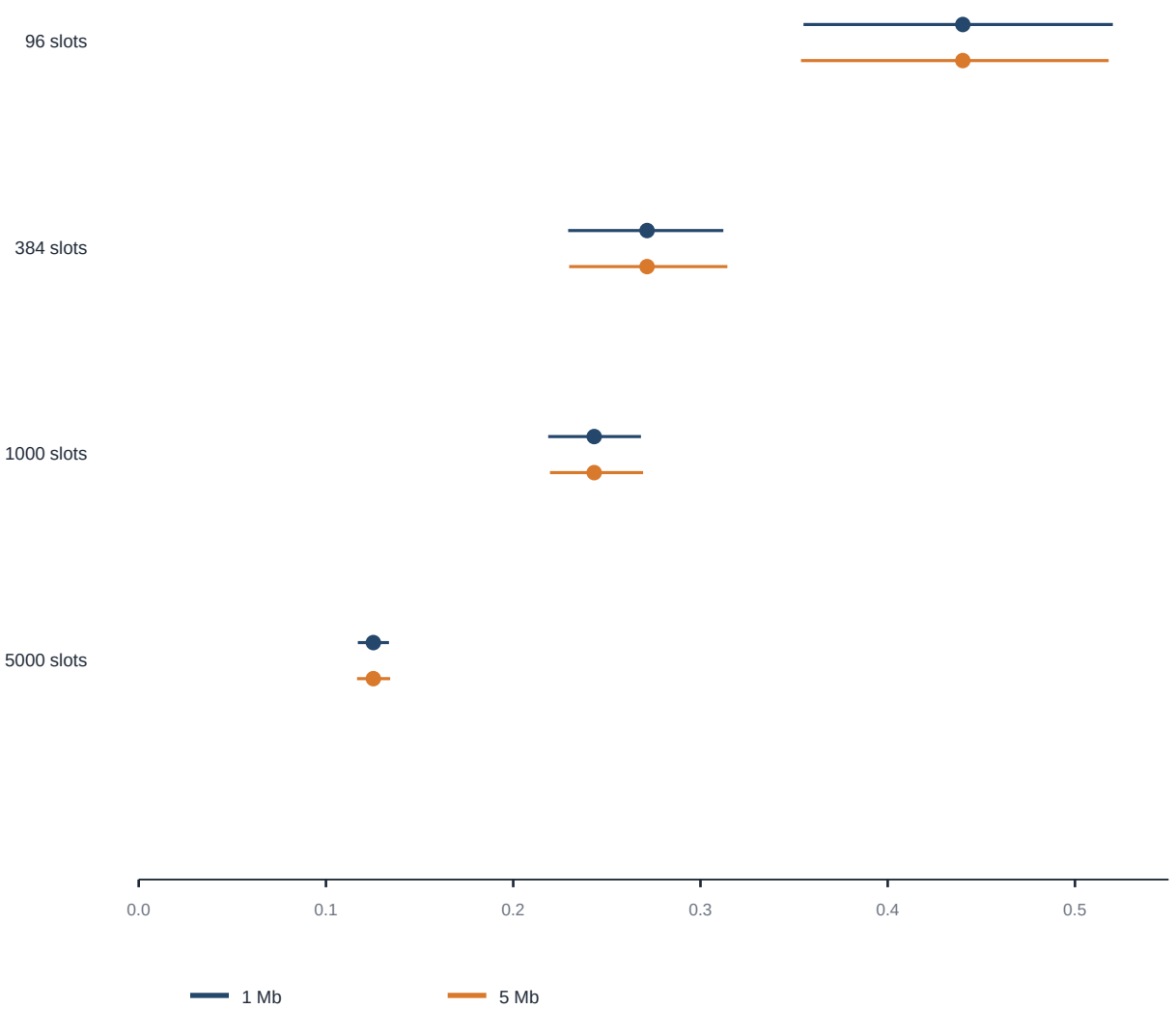


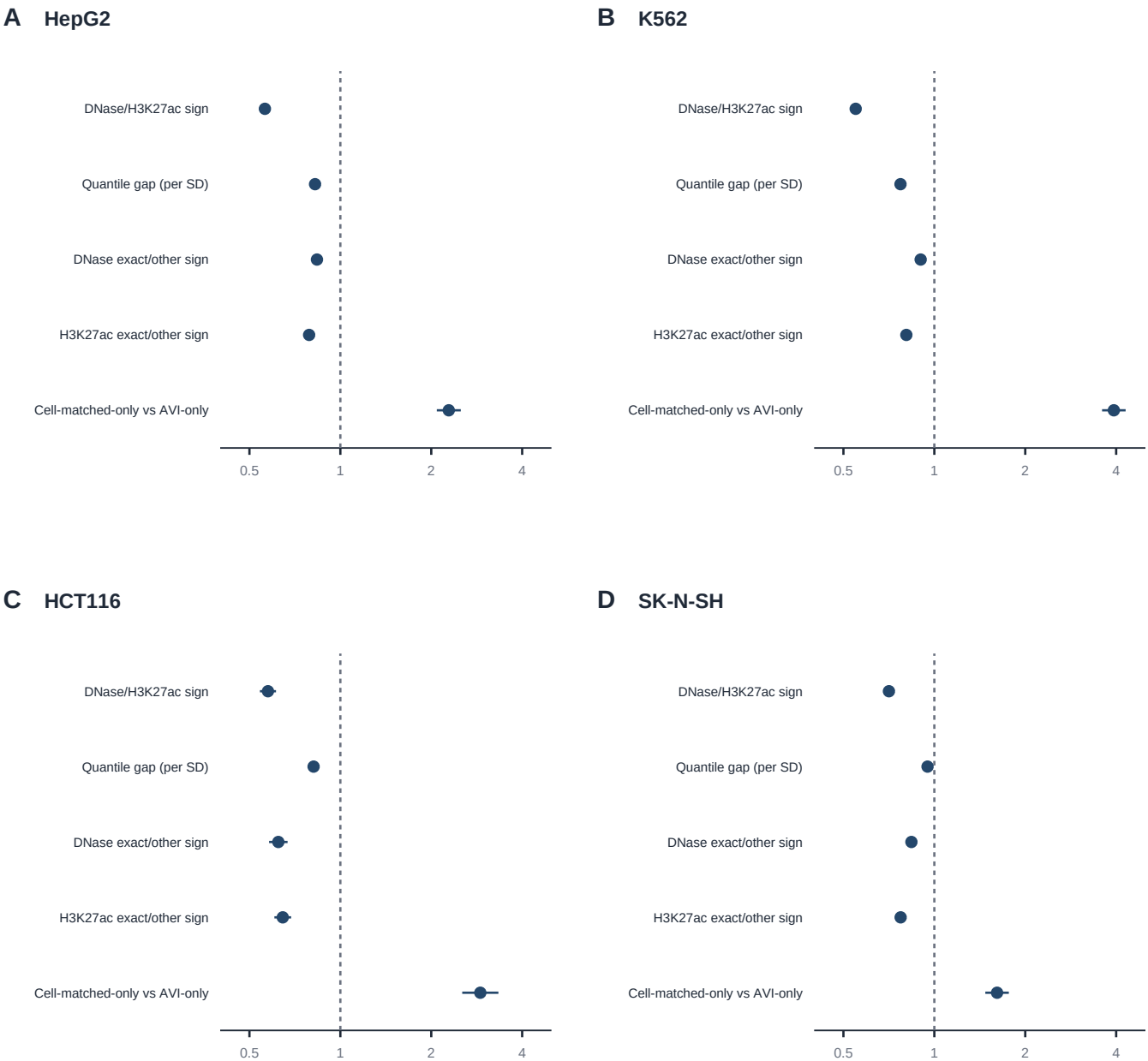
D SK-N-SH



Supplementary Figure S2. Dependence sensitivity

A Cell-matched - AVI precision by block size



Nearly identical intervals show robustness to wider genomic dependence.

Supplementary Figure S3. Prespecified uncertainty indicators